

Insertion Loss & Heat-Load Projections

Hard numbers for RF insertion loss and conducted heat load of two candidate micro-coax cables, at assembly lengths of **3"**, **5"**, **10"**, **15"** and **18"**.

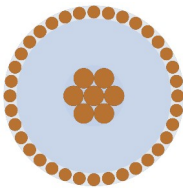
Samtec TCF-3450F			NEW N12-42M-100006-1		
34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω			34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω		
Z ₀		50.5 Ω	Z ₀		49.8 Ω
Cable OD		0.56 mm · 22.1 mil	Cable OD		0.71 mm · 28.0 mil
α @ 2 GHz		0.637 dB/m	α @ 2 GHz		4.39 dB/m

RF model: Pozar conductor loss + dielectric loss, with per-conductor stranding factor K_s calibrated to the TCF-3450F insertion-loss curve (1–4 GHz). Heat load: steady-state conduction across each cryostat stage's ΔT, with lay-angle path corrections. Use ← → to navigate.

3" assembly · 76 mm

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω OD 0.56 mm · 22.1 mil

RF INSERTION LOSS

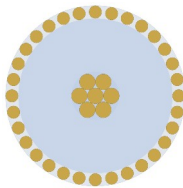
2 GHz	0.049 dB
4 GHz	0.069 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	35.3 mW
4 K → 1 K	566 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

RF INSERTION LOSS

2 GHz	0.335 dB
4 GHz	0.475 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	533 μW
4 K → 1 K	5.1 μW

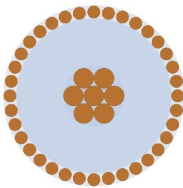
Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

RF insertion loss = total $\alpha(f) \cdot L$ at the 4–40 K stage ($\bar{T} \approx 22$ K), conductor + dielectric. Heat load = steady-state conduction $\dot{Q}$ across each stage's ΔT , per cable. Model: Pozar α_c with per-conductor stranding factor K_s (calibrated to the TCF-3450F IL curve, 1–4 GHz). Both cables 50 Ω.

5" assembly · 127 mm

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω OD 0.56 mm · 22.1 mil

RF INSERTION LOSS

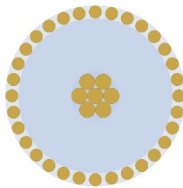
2 GHz	0.081 dB
4 GHz	0.115 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	21.2 mW
4 K → 1 K	339 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

RF INSERTION LOSS

2 GHz	0.558 dB
4 GHz	0.791 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	320 μW
4 K → 1 K	3.1 μW

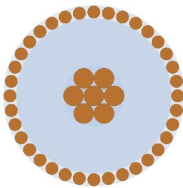
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RF insertion loss = total $\alpha(f) \cdot L$ at the 4–40 K stage ($\bar{T} \approx 22$ K), conductor + dielectric. Heat load = steady-state conduction $\dot{Q}$ across each stage's ΔT , per cable. Model: Pozar α_c with per-conductor stranding factor K_s (calibrated to the TCF-3450F IL curve, 1–4 GHz). Both cables 50 Ω.

10" assembly · 254 mm

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω OD 0.56 mm · 22.1 mil

RF INSERTION LOSS

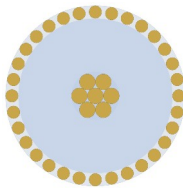
2 GHz	0.162 dB
4 GHz	0.230 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	10.6 mW
4 K → 1 K	170 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

RF INSERTION LOSS

2 GHz	1.12 dB
4 GHz	1.58 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	160 μW
4 K → 1 K	1.5 μW

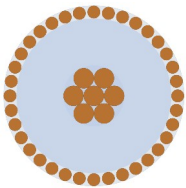
Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

RF insertion loss = total $\alpha(f) \cdot L$ at the 4–40 K stage ($\bar{T} \approx 22$ K), conductor + dielectric. Heat load = steady-state conduction $\dot{Q}$ across each stage's ΔT , per cable. Model: Pozar α_c with per-conductor stranding factor K_s (calibrated to the TCF-3450F IL curve, 1–4 GHz). Both cables 50 Ω.

15" assembly · 381 mm

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω OD 0.56 mm · 22.1 mil

RF INSERTION LOSS

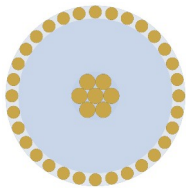
2 GHz	0.243 dB
4 GHz	0.346 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	7.07 mW
4 K → 1 K	113 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

RF INSERTION LOSS

2 GHz	1.67 dB
4 GHz	2.37 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	107 μW
4 K → 1 K	1.0 μW

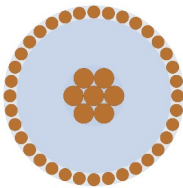
Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

RF insertion loss = total $\alpha(f) \cdot L$ at the 4–40 K stage ($\bar{T} \approx 22$ K), conductor + dielectric. Heat load = steady-state conduction $\dot{Q}$ across each stage's ΔT , per cable. Model: Pozar α_c with per-conductor stranding factor K_s (calibrated to the TCF-3450F IL curve, 1–4 GHz). Both cables 50 Ω.

18" assembly · 457 mm

Samtec TCF-3450F

34 AWG (7×42) SPC / FEP-foam / 46 AWG SPC serve shield, 50 Ω



Z_0 50.5 Ω OD 0.56 mm · 22.1 mil

RF INSERTION LOSS

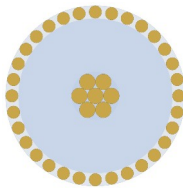
2 GHz	0.291 dB
4 GHz	0.415 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	5.89 mW
4 K → 1 K	94 μW

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

RF INSERTION LOSS

2 GHz	2.01 dB
4 GHz	2.85 dB

CONDUCTED HEAT LOAD · PER CABLE

40 K → 4 K	89 μW
4 K → 1 K	0.8 μW

Trade-off · TCF-3450F 6.9× lower loss · N12-42M 66× lower heat

RF insertion loss = total $\alpha(f) \cdot L$ at the 4–40 K stage ($\bar{T} \approx 22$ K), conductor + dielectric. Heat load = steady-state conduction $\dot{Q}$ across each stage's ΔT , per cable. Model: Pozar α_c with per-conductor stranding factor K_s (calibrated to the TCF-3450F IL curve, 1–4 GHz). Both cables 50 Ω.

All lengths at a glance

Samtec TCF-3450F						NEW N12-42M-100006-1				
Length	LOSS 2 GHz	Q̇ PER CABLE		208 (13 CABLES OF 16 LINES)		LOSS 2 GHz	Q̇ PER CABLE		208 (13 CABLES OF 16 LINES)	
		40→4 K	4→1 K	40→4 K	4→1 K		40→4 K	4→1 K	40→4 K	4→1 K
3"	0.049 dB	35.3 mW	566 μW	7.35 W	117.6 mW	0.335 dB	533 μW	5.1 μW	110.9 mW	1.06 mW
5"	0.081 dB	21.2 mW	339 μW	4.41 W	70.6 mW	0.558 dB	320 μW	3.1 μW	66.6 mW	636 μW
10"	0.162 dB	10.6 mW	170 μW	2.20 W	35.3 mW	1.12 dB	160 μW	1.5 μW	33.3 mW	318 μW
15"	0.243 dB	7.07 mW	113 μW	1.47 W	23.5 mW	1.67 dB	107 μW	1.0 μW	22.2 mW	212 μW
18"	0.291 dB	5.89 mW	94 μW	1.22 W	19.6 mW	2.01 dB	89 μW	0.8 μW	18.5 mW	177 μW

Insertion loss quoted at 2 GHz, 4–40 K stage. Heat load is conducted Q̇ across each stage's ΔT — shown per cable and for the full 208-cable harness (13 × 16). See per-length slides for 4 GHz figures and inner/outer/dielectric breakdowns.

NEW-Doubled vs baseline · 5" assembly · 127 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

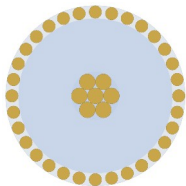
per cable **320** μ W
× 208 harness **66.6** mW

RF INSERTION LOSS

2 GHz **0.558** dB
4 GHz **0.791** dB

NEW-Doubled

2× N12-42M scale — 28 AWG (7×36) brass / PFA / 38 AWG brass spiral shield, 50 Ω



Z_0 49.7 Ω OD 1.42 mm · 55.9 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable **1.29** mW
× 208 harness **267.5** mW

RF INSERTION LOSS

2 GHz **0.281** dB
4 GHz **0.399** dB

Trade-off · **NEW-Doubled** 4.0× the heat · 2.0× lower loss · 2.00× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 40 K → 4 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Doubled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.00× the baseline.

OD 0.71 mm • 28.0 mil

3.1 μW

636 μW

0.553 dB

0.783 dB

OD 1.42 mm • 55.9 mil

12 μW

2.56 mW

0.277 dB

0.393 dB

Heat load = steady-state conduction $\dot{Q}$ across the 4 K $\rightarrow$ 1 K ΔT , per cable and for the full 208-cable harness (13 $\times$ 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Doubled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.00 $\times$ the baseline.

NEW-Doubled vs baseline · 10" assembly · 254 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω

Z₀ 49.8 Ω

OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable

× 208 harness

160 μW

33.3 mW

RF INSERTION LOSS

2 GHz

4 GHz

1.12 dB

1.58 dB



NEW-Doubled

2× N12-42M scale — 28 AWG (7×36) brass / PFA / 38 AWG brass spiral shield, 50 Ω

Z₀ 49.7 Ω

OD 1.42 mm · 55.9 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable

× 208 harness

643 μW

133.8 mW

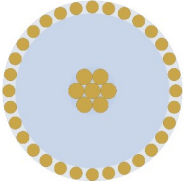
RF INSERTION LOSS

2 GHz

4 GHz

0.561 dB

0.798 dB



Trade-off · NEW-Doubled 4.0× the heat · 2.0× lower loss · 2.00× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 40 K → 4 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Doubled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.00× the baseline.

NEW-Doubled vs baseline · 10" assembly · 254 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

1.5 μW

× 208 harness

318 μW

RF INSERTION LOSS

2 GHz

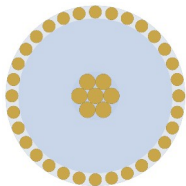
1.11 dB

4 GHz

1.57 dB

NEW-Doubled

2× N12-42M scale — 28 AWG (7×36) brass / PFA / 38 AWG brass spiral shield, 50 Ω



Z_0 49.7 Ω

OD 1.42 mm · 55.9 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

6.1 μW

× 208 harness

1.28 mW

RF INSERTION LOSS

2 GHz

0.555 dB

4 GHz

0.786 dB

Trade-off · NEW-Doubled 4.0× the heat · 2.0× lower loss · 2.00× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 4 K → 1 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Doubled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.00× the baseline.

NEW-Tripled vs baseline · 5" assembly · 127 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

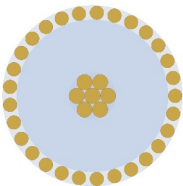
per cable **320** μ W
× 208 harness **66.6** mW

RF INSERTION LOSS

2 GHz **0.558** dB
4 GHz **0.791** dB

NEW-Tripled

≈3× N12-42M scale — 25 AWG (7×33) brass / PFA / 34 AWG brass spiral shield, 50 Ω



Z_0 50.0 Ω OD 2.06 mm · 81.1 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable **2.84** mW
× 208 harness **590.9** mW

RF INSERTION LOSS

2 GHz **0.197** dB
4 GHz **0.281** dB

Trade-off · **NEW-Tripled** 8.9× the heat · 2.8× lower loss · 2.90× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 40 K → 4 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with p , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Tripled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.90× the baseline.

NEW-Tripled vs baseline · 5" assembly · 127 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

3.1 μW

× 208 harness

636 μW

RF INSERTION LOSS

2 GHz

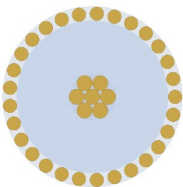
0.553 dB

4 GHz

0.783 dB

NEW-Tripled

≈3× N12-42M scale — 25 AWG (7×33) brass / PFA / 34 AWG brass spiral shield, 50 Ω



Z_0 50.1 Ω

OD 2.06 mm · 81.1 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

27 μW

× 208 harness

5.65 mW

RF INSERTION LOSS

2 GHz

0.194 dB

4 GHz

0.276 dB

Trade-off · NEW-Tripled 8.9× the heat · 2.8× lower loss · 2.90× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 4 K → 1 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Tripled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.90× the baseline.

NEW-Tripled vs baseline · 10'' assembly · 254 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable

160 μW

× 208 harness

33.3 mW

RF INSERTION LOSS

2 GHz

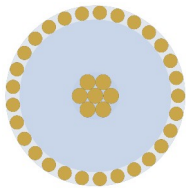
1.12 dB

4 GHz

1.58 dB

NEW-Tripled

≈3× N12-42M scale — 25 AWG (7×33) brass / PFA / 34 AWG brass spiral shield, 50 Ω



Z_0 50.0 Ω

OD 2.06 mm · 81.1 mil

CONDUCTED HEAT LOAD · 40 K → 4 K

per cable

1.42 mW

× 208 harness

295.4 mW

RF INSERTION LOSS

2 GHz

0.394 dB

4 GHz

0.562 dB

Trade-off · NEW-Tripled 8.9× the heat · 2.8× lower loss · 2.90× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 40 K → 4 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with p , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Tripled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.90× the baseline.

NEW-Tripled vs baseline · 10'' assembly · 254 mm

NEW N12-42M-100006-1

34 AWG (7×42) C27000 brass / PFA / 44 AWG brass spiral shield, 50 Ω



Z_0 49.8 Ω

OD 0.71 mm · 28.0 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

1.5 μW

× 208 harness

318 μW

RF INSERTION LOSS

2 GHz

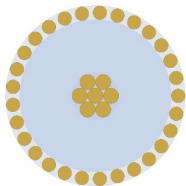
1.11 dB

4 GHz

1.57 dB

NEW-Tripled

≈3× N12-42M scale — 25 AWG (7×33) brass / PFA / 34 AWG brass spiral shield, 50 Ω



Z_0 50.1 Ω

OD 2.06 mm · 81.1 mil

CONDUCTED HEAT LOAD · 4 K → 1 K

per cable

14 μW

× 208 harness

2.82 mW

RF INSERTION LOSS

2 GHz

0.389 dB

4 GHz

0.552 dB

Trade-off · NEW-Tripled 8.9× the heat · 2.8× lower loss · 2.90× the OD

Heat load = steady-state conduction $\dot{Q}$ across the 4 K → 1 K ΔT , per cable and for the full 208-cable harness (13 × 16). RF insertion loss = $\alpha(f) \cdot L$ with ρ , ϵ_r and $\tan \delta$ taken at this stage's $\bar{T}$. NEW-Tripled is the N12-42M construction (brass / PFA / spiral shield) scaled up, re-sized to hold 50 Ω and 0.78 shield coverage; its OD is 2.90× the baseline.